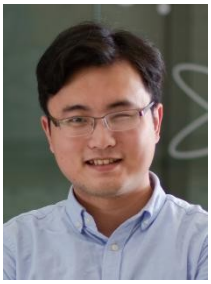
冉仕举 Email: sjran@cnu.edu.cn



个人简介：首都师范大学物理系，教授，博士生导师，现任科研副主任，主要研究量子人工智能、量子多体物理、张量网络理论与方法、量子信息与量子计算；主讲本科生专业课“电动力学”、“数学物理方法”、必修课“人工智能导论 A”；在 PRL 等发表论文 60 余篇，提出的树状幺正张量网络机器学习相关论文获国际高被引奖；以第一作者于 Springer 出版社出版英文专著《Tensor Network Contractions》，获国际著名数学刊物正面评述；独立出版中文著作《张量网络》，获批国家出版基金；获批国家发明专利 4 项；担任国家自然科学基金委青年项目负责人、重点项目主要参与人，北京市自然科学基金委面上项目、教委一般项目负责人中科院专项子任务负责人；PRL、Nature 子刊等审稿人，多次在国际学术会议作邀请报告；曾任 Q1\Q2 区 SCI 期刊客座编辑、机器学习国际会议委员会成员、欧盟学术委员会项目评审人；获北京市海外人才计划、北京市优秀指导教师，任中国计算机学会量子计算专委执行委员。

招生方向：**AI+物理学**。主要招收 AI 与物理学的交叉研究方向，包括 AI 大模型与理论物理（机器推导、数值计算等）、量子计算等的结合，张量网络机器学习方法机器及其解释性理论研究。欢迎对这方面感兴趣、有热情的同学加入研究团队。

教育与研究经历：

2006 年 9 月 - 2010 年 7 月，北京师范大学物理学系，本科；

2010 年 9 月 - 2015 年 7 月，中国科学院大学物理科学学院，博士（导师：苏刚教授）；

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2015 年 7 月 - 2018 年 7 月，西班牙 ICFO，博士后（合作导师：M. Lewenstein 教授）；

2016 年，获 Fundacio-Catalunya 独立博士后研究员 fellowship；

2017 年 9 月，德国马克斯-普朗克量子光学研究所，访问学者；

2018 年 6 月，德国美因茨大学，访问学者；

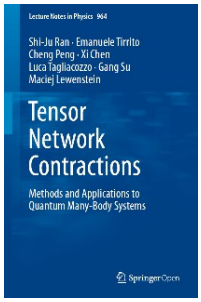
2018 年 7 月-11 月，中国科学院大学，访问学者；

2018 年 11 月 - 2018 年 12 月，首都师范大学物理系，讲师

2019 年 1 月 - 2023 年 12 月，首都师范大学物理系，副研究员

2024 年 1 月 - 今，首都师范大学物理系，教授

学术专著：



- **Shi-Ju Ran***, Emanuele Tirrito, Cheng Peng, Xi Chen, Luca Tagliacozzo, Gang Su, and Maciej Lewenstein, "Tensor Network Contractions", *Lecture Notes in Physics, Springer, Cham* (2020). ISSN: 0075-8450; ISBN: 978-3-030-34488-7. DOI: https://doi.org/10.1007/978-3-030-34489-4

该书系统介绍了张量网络量子物理方法，总结了作者在该领域的一系列创新成果，据出版社官方数据，出版至今全球下载量 10 万余次。



- 冉仕举，《张量网络》，首都师范大学出版社，北京 (2022)，ISBN: 978-7-5656-7150-0

首部张量网络方面的中文专著，针对张量网络在量子物理、应用数学、人工智能等多个领域的高交叉性，从张量的基本定义出发，循序渐进地介绍了量子物理、统计、张量基础，到最新的量子物理与机器学习方法，旨在为物理专业的本科生、研究生、学者，以及在非量子物理领域的学者与读者提供一个系统了解张量网络的途径。该著作获批 2023 年国家出版基金。

学术论文：

[1] Da-zhi Fang, **Shi-Ju Ran***, and Gang Su*, Exotic transition and reentrance on the 1/9

magnetization plateau in a spin-1/2 anisotropic kagome antiferromagnet *Phys. Rev. B* **113**, L140405 (10 April 2026).

[2] Shengchen Bai, Kun Zhang, and **Shi-Ju Ran***, Orthogonality in Quantum-Probabilistic Machine Learning: An Investigation on Multi-qubit Encoding, Intell Comput. 5, 0232 (24 March 2026).

[3] Hao Zhu, Ding-Zu Wang*, **Shi-Ju Ran***, and Guo-Feng Zhang*, Statistics-encoded tensor network approach in disordered quantum many-body spin chains, *Phys. Rev. B* **113**, 085123 (11 February 2026).

[4] Sheng-Chen Bai and **Shi-Ju Ran***, Machine learning of chaotic characteristics in classical nonlinear dynamics using variational quantum circuit, *Chinese Phys. B* **35** 020303 (2026).

[5] Shuo Qi, Wen-Jun Li*, Gang Su*, and **Shi-Ju Ran***, Matrix-Product Entanglement Characterizing the Optimality of State-Preparation Quantum Circuits, *Phys. Rev. Lett.* **136**, 020602 (14 January 2026).

[6] Wenbo Qiao, Peng Zhang*, Jiaming Zhao, and **Shi-Ju Ran**, Unveiling the nature of graphs through quantum graphon learning, *npj Quantum Inf.*, **12**, 8 (8 January 2026).

[7] Rui Hong, Hao-Wei Cui, An-Chun Ji, and **Shi-Ju Ran***, Entanglement scaling and criticality of infinite-size quantum many-body systems in continuous space addressed by a tensor network approach, *Phys. Rev. B* **111**, 245109 (3 June 2025).

[8] Yong Qing, Ke Li, Peng-Fei Zhou, and **Shi-Ju Ran***, Compressing Neural Networks Using Tensor Networks with Exponentially Fewer Variational Parameters, *Intell. Comput.* **4**, 0123 (15 May 2025).

[9] Sheng-Chen Bai and **Shi-Ju Ran***, Universal Scaling Laws in Quantum-Probabilistic Machine Learning by Tensor Network: Toward Interpreting Representation and Generalization Powers, *Chin. Phys. Lett.* **41**, 120701 (19 November 2024).

[10] Ying Lu, Pei Shi, Xiao-Han Wang, Jie Hu, and **Shi-Ju Ran***, Persistent Ballistic Entanglement Spreading with Optimal Control in Quantum Spin Chains, *Phys. Rev. Lett.* **133**, 070402 (13 August 2024).

[11] Ding-Zu Wang, Hao Zhu, Jian Cui, Javier Argüello-Luengo, Maciej Lewenstein, Guo-Feng Zhang*, Piotr Sierant*, and **Shi-Ju Ran***, Eigenstate thermalization and its breakdown in quantum spin chains with inhomogeneous interactions, *Phys. Rev. B* **109**, 045139 (22 January 2024).

[12] Ding-Zu Wang, Guo-Feng Zhang*, Maciej Lewenstein*, and **Shi-Ju Ran***, Boundary-induced singularity in strongly-correlated quantum systems at finite temperature, *Quantum Science and Technology* **9**, 015008 (2024).

[13] **Shi-Ju Ran*** and Gang Su*, Tensor network for interpretable and efficient quantum-inspired machine learning, *Intelligent Computing* **2**, 0061 (17 Nov 2023).

[14] Ying Lu and **Shi-Ju Ran***, Many-body control with reinforcement learning and tensor networks, *Nature Machine Intelligence*, News&Views (4 October 2023).

[15] Peng-Fei Zhou, Ying Lu, Jia-Hao Wang, and **Shi-Ju Ran***, Tensor Network Efficiently Representing Schmidt Decomposition of Quantum Many-Body States, *Phys. Rev. Lett.* **131**, 020403 (13 July 2023).

[16] Da-Zhi Fang, Ning Xi, **Shi-Ju Ran***, and Gang Su*, Nature of the 1/9-magnetization plateau in the spin-1/2 kagome Heisenberg antiferromagnet, *Phys. Rev. B* **107**, L220401 (13 June 2023).

[17] Ye-Ming Meng, Jing Zhang, Peng Zhang, Chao Gao*, and **Shi-Ju Ran***, Residual matrix product state for machine learning, *SciPost Phys.* **14**, 142 (02 June 2023).

[18] Ying Lu, Peng-Fei Zhou, Shao-Ming Fei*, and **Shi-Ju Ran***, Quantum compiling with a variational instruction set for accurate and fast quantum computing, *Phys. Rev. Research* **5**, 023096 (12 May 2023).

[19] 冉仕举、苏刚，张量网络量子机器学习：空间、测度与概率诠释，中国计算机学会通讯，19(4)，17-21，2023。

[20] 张鹏、冉仕举（特邀编辑），量子人工智能，中国计算机学会通讯，19(4)，8-9，2023。

[21] **Shi-Ju Ran***, Qibin Zhao, Peng Zhang, and Chu Guo, Editorial: Tensor network approaches for quantum many-body physics and machine learning, *Frontiers in Physics* **11**, 1170492 (27 February 2023).

[22] Sheng-Chen Bai, Yi-Cheng Tang, and **Shi-Ju Ran***, Unsupervised Recognition of Informative Features via Tensor Network Machine Learning and Quantum Entanglement Variations, *Chinese Phys. Lett.* **39**, 100701 (15 September 2022).

[23] Ding-Zu Wang, Guo-Feng Zhang*, Maciej Lewenstein*, and **Shi-Ju Ran***, Efficient simulation of quantum many-body thermodynamics by tailoring a zero-temperature tensor network, *Phys. Rev. B* **105**, 155155 (27 April 2022).

[24] Rui Hong, Ya-Xuan Xiao, Jie Hu, An-Chun Ji, and **Shi-Ju Ran***, Functional tensor network solving many-body Schrödinger equation, *Phys. Rev. B* **105**, 165116 (12 April 2022).

[25] Wei-Ming Li and **Shi-Ju Ran***, Non-Parametric Semi-Supervised Learning in Many-Body Hilbert Space with Rescaled Logarithmic Fidelity, *Mathematics* **10**, 940 (15 March 2022).

[26] Monika Aidelsburger, Luca Barbiero, Alejandro Bermudez, Titas Chanda, Alexandre Dauphin, Daniel González-Cuadra, Przemysław R. Grzybowski, Simon Hands, Fred Jendrzejewski, Johannes Jünemann, Gediminas Juzeliunas, Valentin Kasper, Angelo Piga, **Shi-Ju Ran**, Matteo Rizzi, Germán Sierra, Luca Tagliacozzo, Emanuele Tirrito, Torsten V. Zache, Jakub Zakrzewski, Erez Zohar, and Maciej Lewenstein*, Cold atoms meet lattice gauge theory, *Phil. Trans. R. Soc. A* **380**, 20210064 (20 December 2021).

[27] Ying Lu, Yue-Min Li, Peng-Fei Zhou, and **Shi-Ju Ran***, Preparation of Many-body Ground States by Time Evolution with Variational Microscopic Magnetic Fields and Incomplete Interactions, *Phys. Rev. A* **104**, 052413 (11 November 2021).

[28] Kunkun Wang, Lei Xiao, Wei Yi*, **Shi-Ju Ran***, and Peng Xue*, Experimental realization of a quantum image classifier via tensor-network-based machine learning, *Photon. Res.* **9**, 2332-2340 (8 November 2021) (Editor's pick).

[29] Peng-Fei Zhou, Rui Hong, and **Shi-Ju Ran***, Automatically differentiable quantum circuit for many-qubit state preparation, *Phys. Rev. A* **104**, 042601 (1 October 2021).

[30] Xinran Ma, Z. C. Tu, and **Shi-Ju Ran***, Deep Learning Quantum States for Hamiltonian Estimation, *Chin. Phys. Lett. (Express Letter)* **38** (11), 110301 (11 October 2021) (封面文章).

[31] Rui Hong, Peng-Fei Zhou, Bin Xi, Jie Hu, An-Chun Ji, and **Shi-Ju Ran***, Predicting

quantum potentials by deep neural network and metropolis sampling, *SciPost Physics Core* **4**, 022 (13 September 2021).

[32] Yuhan Liu, Wen-Jun Li, Xiao Zhang, Maciej Lewenstein, Gang Su*, and **Shi-Ju Ran***, Entanglement-Based Feature Extraction by Tensor Network Machine Learning, *Front. Appl. Math. Stat.* **7**, 716044 (06 August 2021).

[33] Yuan Yang, Zheng-Zhi Sun, **Shi-Ju Ran***, and Gang Su*, Visualizing quantum phases and identifying quantum phase transitions by nonlinear dimensional reduction, *Phys. Rev. B* **103**, 075106 (2 February 2021).

[34] Yuan Yang, Zhengchuan Wang*, **Shi-Ju Ran***, and Gang Su*, Phase identification in many-body systems by virtual configuration binarization, *Phys. Rev. E* **103**, 013313 (22 January 2021).

[35] **Shi-Ju Ran***, Zheng-Zhi Sun, Shao-Ming Fei, Gang Su, and Maciej Lewenstein, Tensor network compressed sensing with unsupervised machine learning, *Phys. Rev. Research* **2**, 033293 (24 August 2020).

[36] Zheng-Zhi Sun, **Shi-Ju Ran*,** and Gang Su*, Tangent-Space Gradient Optimization of Tensor Network for Machine Learning, *Phys. Rev. E* **102**, 012152 (30 July 2020).

[37] **Shi-Ju Ran**, Encoding of matrix product states into quantum circuits of one- and two-qubit gates, *Phys. Rev. A* **101**, 032310 (9 March 2020).

[38] Zheng-Zhi Sun, Cheng Peng, Ding Liu, **Shi-Ju Ran***, and Gang Su*, Generative Tensor Network Classification Model for Supervised Machine Learning, *Phys. Rev. B* **101**, 075135 (25 February 2020).

[39] Yuan Yang, **Shi-Ju Ran***, Xi Chen, Zheng-zhi Sun, Shou-Shu Gong, Zhengchuan Wang*, and Gang Su*, Reentrance of Topological Phase in Spin-1 Frustrated Heisenberg Chain, *Phys. Rev. B* **101**, 045133, (29 January 2020).

[40] **Shi-Ju Ran***, Bin Xi, Cheng Peng, Gang Su, and Maciej Lewenstein, Efficient quantum simulation for thermodynamics of infinite-size many-body systems in arbitrary dimensions, *Phys. Rev. B* **99**, 205132 (20 May 2019).

[41] Ding Liu, **Shi-Ju Ran***, Peter Wittek*, Cheng Peng, Rual Blázquez Garca, Gang Su, and Maciej Lewenstein, Machine Learning by Unitary Tensor Network of Hierarchical Tree Structure, *New J. Phys.* **21**, 073059 (30 July 2019).

[42] Xi Chen, **Shi-Ju Ran***, Shuo Yang, Maciej Lewenstein, Gang Su*, Noise-tolerant Signature of Z_N Topological Orders in Quantum Many-body States, *Phys. Rev. B* **99**, 195101 (1 May 2019).

[43] 彭程、冉仕举，时间矩阵乘积态理论及其应用，中国科学院大学学报，36(6)，2019。

[44] **Shi-Ju Ran***, Cheng Peng, Gang Su, and Maciej Lewenstein, Controlling the phase diagram of finite spin-1/2 chains by tuning the boundary interactions, *Phys. Rev. B* **98**, 085111 (7 August 2018).

[45] **Shi-Ju Ran**, Wei Li, Shou-Shu Gong, Andreas Weichselbaum, Jan von Delft, and Gang Su*, Emergent spin-1 trimerized valence bond crystal in the spin-1/2 Heisenberg model on the star lattice, *Phys. Rev. B* **97**, 075146 (26 February 2018).

[46] Cheng Peng, **Shi-Ju Ran***, Maciej Lewenstein, and Gang Su*, Exotic entanglement

scaling of Heisenberg antiferromagnet on honeycomb lattice, *Eur. Phys. J. B* **91**, 258 (22 October 2018).

[47] Xi Chen, **Shi-Ju Ran**, Tao Liu, Cheng Peng, Yi-Zhen Huang, and Gang Su*, Thermodynamics of spin-1/2 Kagomé Heisenberg antiferromagnet: algebraic paramagnetic liquid and finite-temperature phase diagram, *Sci. Bull.* **63**, 1545–1550 (22 November 2018).

[48] **Shi-Ju Ran***, Angelo Piga, Cheng Peng, Gang Su, and Maciej Lewenstein, Few-body systems capture many-body physics: Tensor network approach, *Phys. Rev. B* **96**, 155120 (13 October 2017).

[49] **Shi-Ju Ran**, Cheng Peng, Wei Li, Maciej Lewenstein, and Gang Su*, Criticality in two-dimensional quantum systems: Tensor network approach, *Phys. Rev. B* **95**, 155114 (10 April 2017).

[50] J. Jünemann, A. Piga, **Shi-Ju Ran**, M. Lewenstein, M. Rizzi, and A. Bermudez*, Exploring Interacting Topological Insulators with Ultracold Atoms: the Synthetic Creutz-Hubbard Model, *Phys. Rev. X* **7**, 031057 (27 September 2017).

[51] Cheng Peng, **Shi-Ju Ran**, Tao Liu, Xi Chen, and Gang Su*, Fermionic algebraic quantum spin liquid in an octa-kagome frustrated antiferromagnet, *Phys. Rev. B* **95**, 075140 (22 February 2017).

[52] Emanuele Tirrito, **Shi-Ju Ran**, Andrew J Ferris, Ian P McCulloch, and Maciej Lewenstein*, Efficient perturbation theory to improve the density matrix renormalization group, *Phys. Rev. B* **95**, 064110 (21 February 2017).

[53] **Shi-Ju Ran**, Ab-initio optimization principle for the ground states of translationally invariant strongly correlated quantum lattice models, *Phys. Rev. E* **93**, 053310 (27 May 2016).

[54] Meng Wang, **Shi-Ju Ran**, Tao Liu, Yang Zhao, Qing-Rong Zheng, and Gang Su*, Phase diagram and exotic spin-spin correlations of anisotropic Ising model on the Sierpiński gasket, *Eur. Phys. J. B* **89**, 1-10 (1 February 2016).

[55] Tao Liu, **Shi-Ju Ran**, Wei Li, Xin Yan, Yang Zhao, and Gang Su*, Featureless quantum spin liquid, 1/3-magnetization plateau state, and exotic thermodynamic properties of the spin-1/2 frustrated Heisenberg antiferromagnet on an infinite Husimi lattice, *Phys. Rev. B* **89**, 054426 (24 February 2014).

[56] **Shi-Ju Ran**, Bin Xi, Tao Liu, and Gang Su*, Theory of network contractor dynamics for exploring thermodynamic properties of two-dimensional quantum lattice models, *Phys. Rev. B* **88**, 064407 (12 August 2013).

[57] Yang Zhao, Wei Li, Bin Xi, Zhe Zhang, Xin Yan, **Shi-Ju Ran**, Tao Liu, and Gang Su*, Kosterlitz-Thouless phase transition and re-entrance in an anisotropic three-state Potts model on the generalized kagome lattice, *Phys. Rev. E* **87**, 032151 (22 March 2013).

[58] Yang Zhao, Wei Li, Bin Xi, **Shi-Ju Ran**, Yuan-Yuan Zhu, Bing-Wu Wang, Song Gao, and Gang Su*, Honeycomb Heisenberg spin ladder: Unusual ground state and thermodynamic properties, *Eur. Phys. Lett.* **104**, 57009 (23 December 2013).

[59] **Shi-Ju Ran**, Wei Li, Bin Xi, Zhe Zhang, and Gang Su*, Optimized decimation of tensor networks with super-orthogonalization for two-dimensional quantum lattice models, *Phys. Rev. B* **86**, 134429 (26 October 2012).

[60] Xin Yan, Wei Li, Yang Zhao, **Shi-Ju Ran**, and Gang Su*, Phase diagrams, distinct conformal anomalies, and thermodynamics of spin-1 bond-alternating Heisenberg antiferromagnetic chain in magnetic fields, *Phys. Rev. B* **85**, 134425 (13 April 2012).

[61] Wei Li, **Shi-Ju Ran**, Shou-Shu Gong, Yang Zhao, Bin Xi, Fei Ye, and Gang Su*, Linearized tensor renormalization group algorithm for the calculation of thermodynamic properties of quantum lattice models, *Phys. Rev. Lett.* **106**, 127202 (22 March 2011).

[62] Wei Li, Shou-Shu Gong, Yang Zhao, **Shi-Ju Ran**, Song Gao, and Gang Su*, Phase transitions and thermodynamics of the two-dimensional Ising model on a distorted kagome lattice, *Phys. Rev. B* **82**, 134434 (21 October 2010).

发明专利

- 国家发明专利：神经网络的压缩方法、装置、电子设备及存储介质；专利号：ZL 2023 1 0460357.7；发明人：冉仕举，卿勇，李珂，周鹏飞；专利授权国：中华人民共和国；授权公告日：2023 年 7 月 14 日
- 国家发明专利：邮件处理方法、装置、电子设备及存储介质；专利号：ZL 2025 1 0765361.3；发明人：冉仕举，白生辰；专利授权国：中华人民共和国；授权公告日：2025 年 9 月 16 日
- 国家发明专利：基于张量结构优化人工智能模型的方法；专利号：ZL 2026 1 0416463.9；发明人：苏刚，卢迎，冉仕举，周鹏飞；专利授权国：中华人民共和国；授权公告日：2026 年 6 月 30 日
- 国家发明专利：心电图识别方法、装置、电子设备及存储介质；专利号：ZL 2026 1 0087605.1；发明人：冉仕举，张琨，白生辰，丁蕾，唐闽；专利授权国：中华人民共和国；授权公告日：2026 年 9 月 8 日

主办学术会议

- 2026 年 8 月 23 日至 28 日，举办第三届量子智能计算研讨会暨首届“光量青成”量子智能计算暑期学校，地址：四川省都江堰市，主办单位：量子物理与光量子信息教育部重点实验室、首都师范大学、天津大学，组委会联络人：王子竹、冉仕举、张鹏
- 2025-1-10 至 2025-1-12, 举办第二届量子智能计算研讨会, 地址：四川省都江堰市, 主办单位：电子科技大学、首都师范大学、天津大学, 组委会：王子竹、冉仕举、张鹏
- 2020-11-21 至 2020-11-22, 举办量子物理与智能计算交叉研讨会, 地址：线上会议，主办单位：天津大学、首都师范大学,主委会： 张鹏，冉仕举

邀请报告与会议论文（2020 年至今）：

[1] 中文报告：冉仕举，Quantum Inspired optimization methods: for Quantum Many-body Problems，2026 年人工智能与量子科技冬季学校，深圳河套学院，深圳（2026-1-22）

[2] 中文报告：冉仕举，基于张量网络的量子启发机器学习，“LLM and Science“ Program，中国科学院理论物理研究所，北京（2024-12-12）

[3] 英文报告：**Shi-Ju Ran**，Quantum-Inspired Machine Learning Based on Tensor

Networks，The 1st Quantum Intelligence Workshop，中国科学院物理所，北京（2025-12-01）

[4] 中文报告：冉仕举，机器学习与量子计算交叉下的张量网络量子多体新方法，第 204 次科学与技术前沿论坛 人工智能的科学基础，中国科学院学部科学与技术前沿论坛，北京（2025-11-30）

[5] 中文报告：冉仕举，Quantum many-body approaches based on variational quantum circuit and tensor network，第二十二届全国凝聚态理论与统计物理学术会议，华南师范大学（2025-11-08）

[6] 中文报告：冉仕举，基于张量网络的量子启发机器学习方法，天津工业大学（2025-10-17）

[7] 中文报告：冉仕举，Quantum many-body approaches based on variational quantum circuit and tensor network，Chongqing-Chengdu Quantum Workshop（2025-08-25）

[8] 中文报告：冉仕举，基于张量网络的量子启发机器学习，CQCC2025 量子人工智能论坛，电子科技大学，成都（2025-7-21）

[9] 会议论文：Sheng-Chen Bai, Kun Zhang, and Shi-Ju Ran, Quantum-Probabilistic Machine Learning: From Catastrophe of Orthogonality to Multi-qubit Encoding, CQCC 2025, CCIS 2733, 26-39, Springer (2026).

[10] 中文报告：冉仕举，张量网络机器学习方法及其应用，“凝聚态物理、计算物理及新兴交叉”研讨会暨暑期讲习班，河南师范大学，新乡（2024-8-23）

[11] 中文报告：冉仕举，强关联量子系统的机器学习数值新方法，全国光电科学与工程学术会议，电子科技大学，成都（2024-7-27）

[12] 中文报告：冉仕举，基于张量网络与机器学习的强关联量子系统数值计算新方法，第 18 届全国磁学理论会议，四川师范大学，成都（2024-06-09）

[13] 中文报告：冉仕举，Machine Learning by Tensor Network and Quantum Computing，“应物之邀，探理之趣”强基论坛暨微纳光子学学术交流报告（十五），北京航空航天大学，北京（2024-5-22）

[14] 中文报告：冉仕举，Persistent ballistic entanglement spreading with optimal control in quantum spin chains，理论物理前沿交叉研讨会，南京师范大学，南京（2024-4-28）

[15] 英文报告：**Shi-Ju Ran**，Persistent ballistic entanglement spreading with optimal control in quantum spin chains，12th Workshop on Quantum Many-Body Computation（2024-4-13）

[16] 中文报告：冉仕举，Quantum-inspired Machine Learning by Tensor Network，QuACT 学术前沿讲座，中国科学院计算所，北京（2023）

[17] 中文报告：冉仕举，Tensor network for efficient and interpretable machine learning，物理学秋季年会，宁夏大学，银川（2023）

[18] 中文报告：冉仕举，张量网络收缩，暑期多体计算讲习班，闽江学院，福州

（2023）

[19] 中文报告：冉仕举，Tensor network for efficient and interpretable machine learning，量子智能和量子材料研讨会，鲁东大学，烟台（2023）

[20] 英文报告：**Shi-Ju Ran**, Tensor network for efficient quantum computation and many-body simulation，Chengdu-Chongqing Quantum Workshop - Quantum Sensing: Theory to Applications，ChengDu，China（2023）

[21] 中文报告：冉仕举，Efficient simulation of frustrated quantum spin systems by Schmidt tensor network state ansatz，第十七届全国磁学理论会议，河北师范大学，河北（2023）

[22] 英文报告：**Shi-Ju Ran**, Efficient Quantum Computing by Optimal Control of Quantum Many-body Dynamics, The 11th workshop on Quantum Many-Body Computation, Fuzhou, Fujian (2023).

[23] 中文报告：冉仕举, Functional Tensor Network Solving Many-body Schrödinger Equation,中国机器学习与应用科学大会，北京 (2022).

[24] 英文报告：**Shi-Ju Ran**, Functional Tensor Network Solving Many-body Schrödinger Equation, The 10th Workshop on Quantum Many-Body Computation, Jiangsu (2022).

[25] 中文报告：冉仕举，经典、量子，各司其职?, CCF 量子计算精英大会，北京 (2022).

[26] 英文报告：**Shi-Ju Ran**, Deep Learning Quantum States For Hamiltonian Predictions, Workshop II: Tensor Network States and Applications (online), IPAM, USA (2021).

[27] 会议论文：Wen-Jun Li, Zheng-Zhi Sun, Ya-Ru Wang, **Shi-Ju Ran***, and Gang Su*, *Matrix product state for quantum-inspired feature extraction and compressed sensing*, Second Workshop on Quantum Tensor Networks in Machine Learning, 35th Conference on Neural Information Processing Systems (2021).

[28] 英文报告：**Shi-Ju Ran**, Simulating Quantum Many-body Systems by Few-body Models at Zero and Finite Temperatures, Workshop on Tensor Networks in Many Body and Lattice Field, TDLI, SJTU, Shanghai (2021).

[29] 中文报告：冉仕举, Deep Learning Quantum States For Hamiltonian Predictions 第十六届全国磁学会议，扬州大学 (2021).

[30] 中文报告：冉仕举，量子多体物理中的机器学习新方法，凝聚态理论前沿暑期讲习班，河北师范大学 (2021).

[31] 中文报告：冉仕举，量子多体物理中的机器学习新方法，YOCSEF 论坛，天津 (2021).

[32] 中文报告：冉仕举，量子物理与机器学习交融下的新方法，第三届学术交流月，北京航空航天大学 (2021).

[33] 会议论文：Zheng-Zhi Sun, **Shi-Ju Ran***, and Gang Su*, *Tangent-space gradient optimization: an efficient update scheme for tensor network machine learning and beyond*, First Workshop on Quantum Tensor Networks in Machine Learning, 34th Conference on Neural Information Processing Systems (2020).

[34] 会议论文：Ding Liu, Zheng-Zhi Sun, Cheng Peng, Gang Su, and **Shi-Ju Ran***, *Generative Tensor Network Classification for Supervised Learning*, International Workshop on

Tensor Network Representations in Machine Learning, the 29th International Joint Conference on Artificial Intelligence (2020).

[35] 中文报告：冉仕举，Bayesian tensor network: towards efficient and interpretable probabilistic machine learning，第一届量子物理与智能计算研讨会(online)，天津-北京 (2020).

基金项目：

(1) 国家自然科学基金委员会，青年科学基金项目, 12004266，基于张量网络与纠缠理论的高效可解释量子机器学习算法, 2021-01-01 至 2023-12-31，结题，主持。

(2) 国家自然科学基金委员会，重点项目, 11834014，量子多体计算新方法发展及其应用，2019-01-01 至 2023-12-31，结题，主要参与人。

(3) 北京市自然科学基金委，面上项目, 1232025，基于张量网络机器学习方法的量子多体态制备与层析，2023-01 至 2025-12，结题，主持。

(4) 北京市自然科学基金委，面上项目, 1192005，基于张量网络方法的量子自旋液体模拟研究，2019-01 至 2021-12，结题，主持。

(5) 北京市教育委员会科技计划，一般项目，KM202010028013，基于张量网络的量子机器学习算法研究，2020-01-01 至 2022-12-31，结题，主持。

人才培养：

（1）已指导毕业 5 名硕士研究生；